**Goal:** Be able to solve (very basic) differential equations by integration.

**Definition.** For $y = f(x)$, a **differential equation** is an equation involving $x$, $y$, and the derivatives of $y$.

Here are some examples ($m$ and $k$ are constants):

| Equation | English Translation |
|---|---|
| $\frac{d^2y}{dx^2} = -9.81$ | "Acceleration due to gravity is constant." |
| $\frac{dy}{dx} = ky$ | "The rate a population grows is proportional to the amount of organisms in the population." |
| $\frac{d[y]}{dt} = -k[x_{(t)}][y_{(t)}]$ | "The speed of a chemical reaction is proportional to the number of interactions between the molecules." |
| $m\frac{d^2y}{dx^2} + k\sin y = 0$ | "The net force acting on an oscillating pendulum is 0." |

$2x+1=5$

**Example 1.** Verify $y = 3e^{2x}$ is a solution to the differential equation $y' = 2y$.

$y' = [3e^{2x}]' = 3e^{2x}\cdot 2 = 6e^{2x}$

$2y = 2\cdot 3e^{2x} = 6e^{2x}$

$\int \frac{dy}{dx}\,dx = \int dy = y$

**Example 2.** Solve the differential equation $y' = \frac{1}{x}$.

$2x+1=5$

$-1 \quad -1$

$\frac{2x}{2} = \frac{4}{2}$

$x = 2$

$y' = \frac{1}{x}$

$\int y'(x)\,dx = \int \frac{1}{x}\,dx$

$y = \ln|x| + C$

**Definition.** The **general solution** to a differential equation is a function that describes all possible solutions of the differential equation.

**Example 3.** Find the general solution to $y' = 3x^2$

$y' = 3x^2$

$\int y' dx = \int 3x^2 dx$

$y = 3\int x^2 dx$

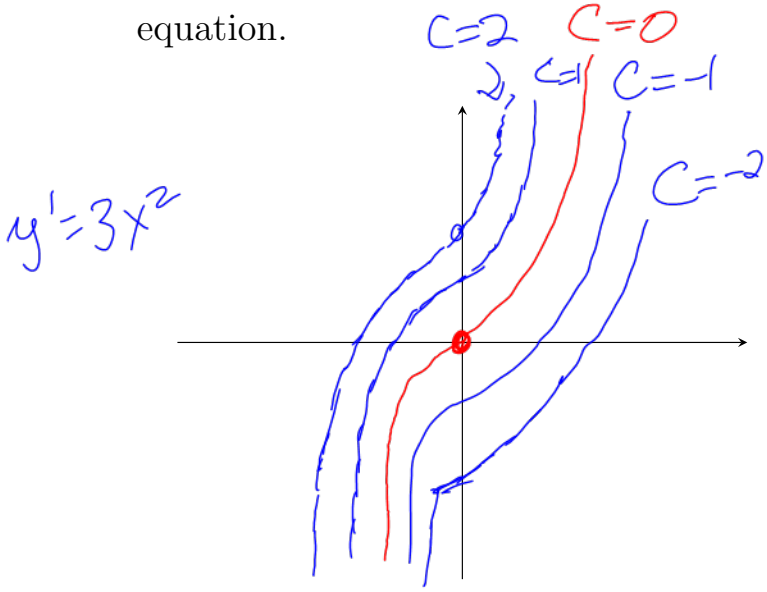
$y = 3 \cdot \frac{x^{2+1}}{2+1} + C$

$y = x^3 + C$

$y' = f(x)$

$y = \int f(x) dx$

The general solution gives us a whole family of solutions to our differential equation.




**Question:** Is there a way to have a unique (i.e., only 1) solution?

**Answer:**

Ask that y also passes through a point (e.g. y(0)=0)

**Definition.** The extra condition above is an example of an initial condition. Usually, we specify when x=0 (but that is not required).

**IVPs**

**Definition.** An **initial value problem (IVP)** is a differential equation combined with initial conditions.

**Example 4.** The system

$$\begin{cases} y' = \sqrt{x} \\ y(0) = 1 \end{cases}$$

is an IVP. How can we solve it?

Step1 Finding the general solution

$$y' = \sqrt{x}$$

$$\int y' \, dx = \int \sqrt{x} \, dx = \int x^{\frac{1}{2}} \, dx$$

$$y(x) = \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C$$

$$y(x) = \frac{2}{3} x^{\frac{3}{2}} + C$$

Step 2 Use the IC to solve for C

Should $y(0) = \frac{2}{3} \cdot 0^{\frac{3}{2}} + C = C$

$$1 = C$$

Conclusion: The solution to the IVP

$$y(x) = \frac{2}{3} x^{\frac{3}{2}} + 1$$

$y = kx$ Proportionality Constant

**Definition.** We say $y$ is **(directly) proportional to** $x$ if there is a constant $k$ such that $y = kx$.

**Example 5.** The size of a population is governed by

$$\frac{dP}{dt} = \sqrt[3]{t}$$

$\frac{dP}{dt}$ is proportional to $\sqrt[3]{t}$ where $k = 1$

where $P$ is the population after $t$ days (i.e., $\frac{dP}{dt}$ is proportional to $\sqrt[3]{t}$ with a constant coefficient of 1). If the initial population is 1000, find the population size after 8 days.

Need to solve

$$\begin{cases} \frac{dP}{dt} = \sqrt[3]{t} \\ P(0) = 1000 \end{cases}$$

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$$\frac{dP}{dt} = \sqrt[3]{t}$$

$$\int \frac{dP}{dt}\,dt = \int t^{\frac{1}{3}}\,dt$$

$$P(t) = \frac{t^{\frac{1}{3}+1}}{\frac{1}{3}+1} + C$$

$$P(t) = \frac{3}{4}t^{\frac{4}{3}} + C$$

$P(0) = 1000$:

1000 Should $P(0) = \frac{3}{4}\cdot 0^{\frac{4}{3}} + C$

$C = 1000$

$\Rightarrow$ $\boxed{P(t) = \frac{3}{4}t^{\frac{4}{3}} + 1000}$

$$P(8) = \frac{3}{4}\left(\sqrt[3]{8}\right)^4 + 1000 = \frac{3}{4}\cdot 16 + 1000 = 12 + 1000$$

$= 1012$ people

**Multiple ICs**

**Fact:** The have a unique solution, you need $N$ ICs where $N$ is the highest order of derivatives in the differential equation.

**Example 6.** Solve the IVP:

$$\begin{cases} y''(t) = t - 3 \\ y'(1) = \frac{1}{2} \\ y(0) = 1 \end{cases}$$




$y''(t) = t - 3$

$\int y''(t)\,dt = \int (t-3)\,dt$

$y'(t) = \frac{1}{2}t^2 - 3t + C$

$y'(1) = \frac{1}{2}$

$\frac{1}{2}$ Should $y'(1) = \frac{1}{2} - 3 + C$

$\frac{1}{2} = \frac{1}{2} - 3 + C$

$0 = -3 + C$

$C = 3$

$\Rightarrow y'(t) = \frac{1}{2}t^2 - 3t + 3$

$\int y'(t)\,dt = \int \left(\frac{1}{2}t^2 - 3t + 3\right) dt$

$y(t) = \frac{1}{2} \cdot \frac{1}{3}t^3 - \frac{3}{2}t^2 + 3t + D$

$y(t) = \frac{1}{6}t^3 - \frac{3}{2}t^2 + 3t + D$

$y(0) = 1$

Should $y(0) = 0 - 0 + 0 + D$

$D = 1$

$\Rightarrow y(t) = \frac{1}{6}t^3 - \frac{3}{2}t^2 + 3t + D$

**Example 7.** A ball is kicked off the ground with an initial velocity of $20\,\mathrm{m\,s^{-1}}$. Describe the height of the ball as a function of time (assume gravity is the only force acting on the ball once it is kicked).

$32\ ft/s^2$

Need to solve the IVP

$$\begin{cases} a(t) = -9.81 \\ v(0) = 20 \\ s(0) = 0 \end{cases}$$

$-\frac{1}{2}gt^2 + [v(0)]t + s(0)$

Position $\xrightarrow{d/dt}$ Velocity $\xrightarrow{d/dt}$ Acceleration

Position $\xleftarrow{\int}$ Velocity $\xleftarrow{\int}$ Acceleration

$$\int a(t)\,dt = \int -9.81\,dt$$

$$v(t) = -9.81t + C$$

$-\frac{1}{2}g$

$$20 = v(0) = C$$

$$\Rightarrow \quad v(t) = -9.81t + 20$$

$$\int v(t)\,dt = \int (-9.81t + 20)\,dt$$

$$s(t) = -4.905t^2 + 20t + C$$

$$0 = s(0) = C$$

**Example 8** (HW 27, Q6)**.** Solve the IVP:

$$\begin{cases} y''(x) = 4\sin x + \frac{10}{\pi^2}x \\ y'(\pi) = 4 \\ y\left(\frac{\pi}{2}\right) = -4 \end{cases}$$

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$$y''(x) = 4\sin x + \frac{10}{\pi^2}x$$

$$\int y''(x)\,dx = \int \left(4\sin x + \frac{10}{\pi^2}x\right)dx$$

$$y'(x) = 4(-\cos x) + \frac{10}{\pi^2}\cdot\frac{1}{2}x^2 + C$$

$$y'(x) = -4\cos x + \frac{5}{\pi^2}x^2 + C$$

$y'(\pi) = 4$

4 should $y'(\pi) = -4\cos\pi + \frac{5}{\pi^2}(\pi^2) + C$

$$4 = 4 + 5 + C$$

$$0 = 5 + C$$

$$C = -5$$

$\Rightarrow$

$$y'(x) = -4\cos x + \frac{5}{\pi^2}x^2 - 5$$

$$\int y'(x)\,dx = \int \left(-4\cos x + \frac{5}{\pi^2}x^2 - 5\right)dx$$

$$y(x) = -4\sin x + \frac{5}{\pi^2}\cdot\frac{1}{3}x^3 - 5x + D$$

$y\left(\frac{\pi}{2}\right) = -4$

$-4$ should $= y\left(\frac{\pi}{2}\right) = -4\sin\left(\frac{\pi}{2}\right) + \frac{5}{3\pi^2}\left(\frac{\pi}{2}\right)^3 - 5\left(\frac{\pi}{2}\right) + D$

$$-4 = -4 + \frac{5}{24}\pi - \frac{5}{2}\pi + D$$

$$D = \frac{5}{2}\pi - \frac{5}{24}\pi = \frac{55}{24}\pi$$

$$y(x) = -4\sin x + \frac{5}{3\pi^2}x^3 - 5x + \frac{55\pi}{24}$$